



Lecture 3.2.0

Introduction to keypoint features

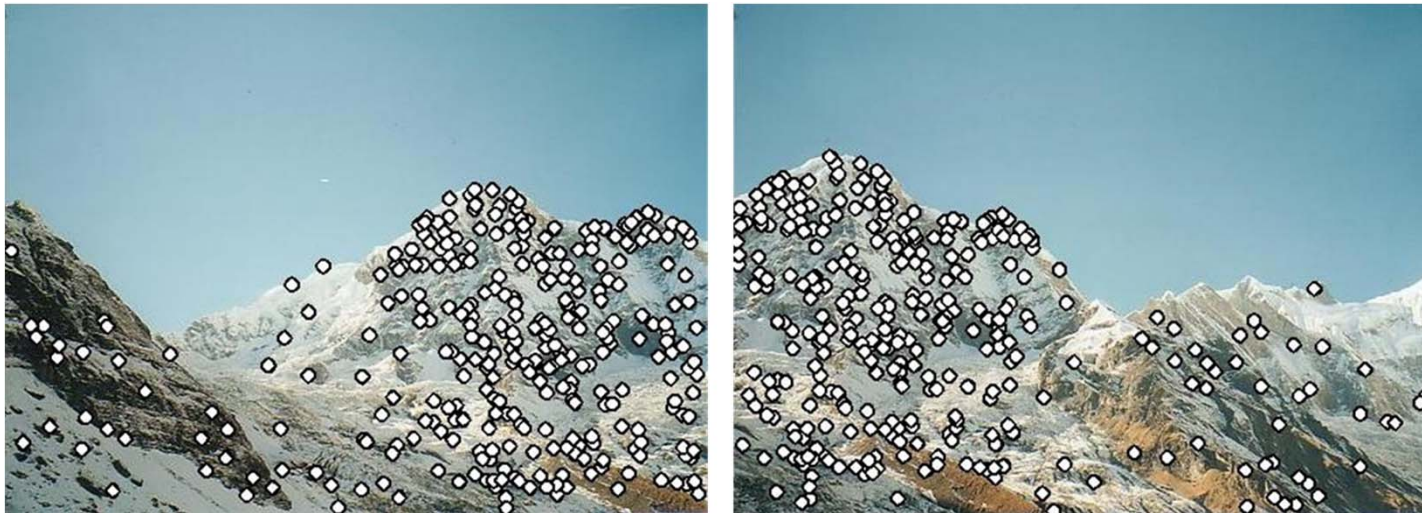
Why extract features?

- Example: Panorama stitching
 - How do we combine two images?



Why extract features?

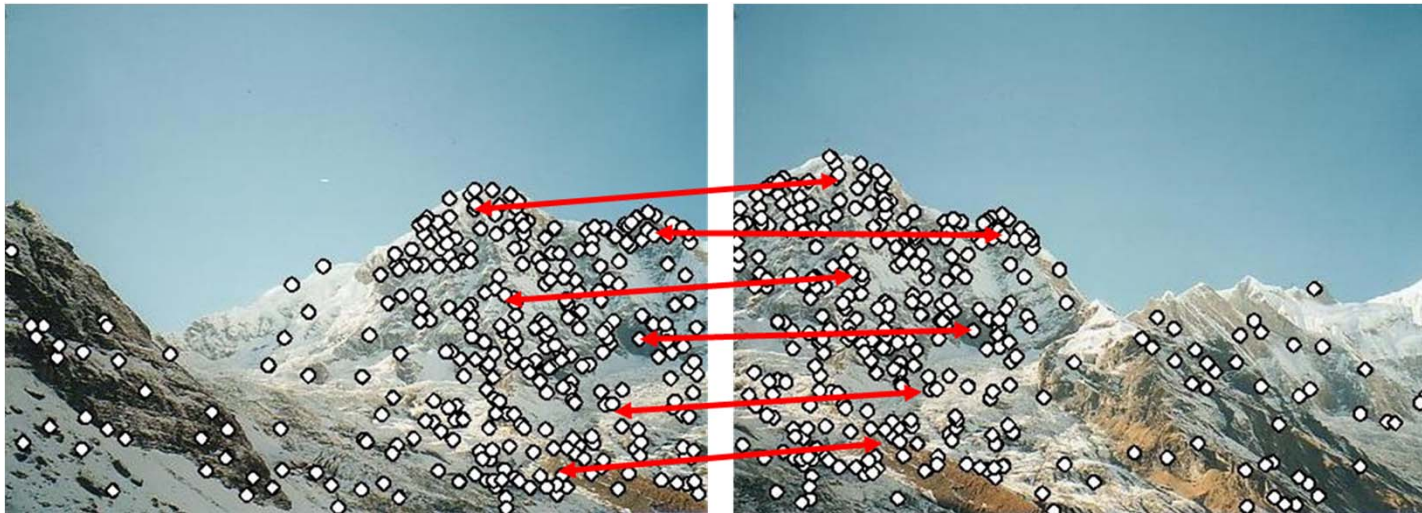
- Example: Panorama stitching
 - How do we combine two images?



Step 1: Extract features

Why extract features?

- Example: Panorama stitching
 - How do we combine two images?



Step 1: Extract features

Step 2: Match features

Why extract features?

- Example: Panorama stitching
 - How do we combine two images?



Step 1: Extract features

Step 2: Match features

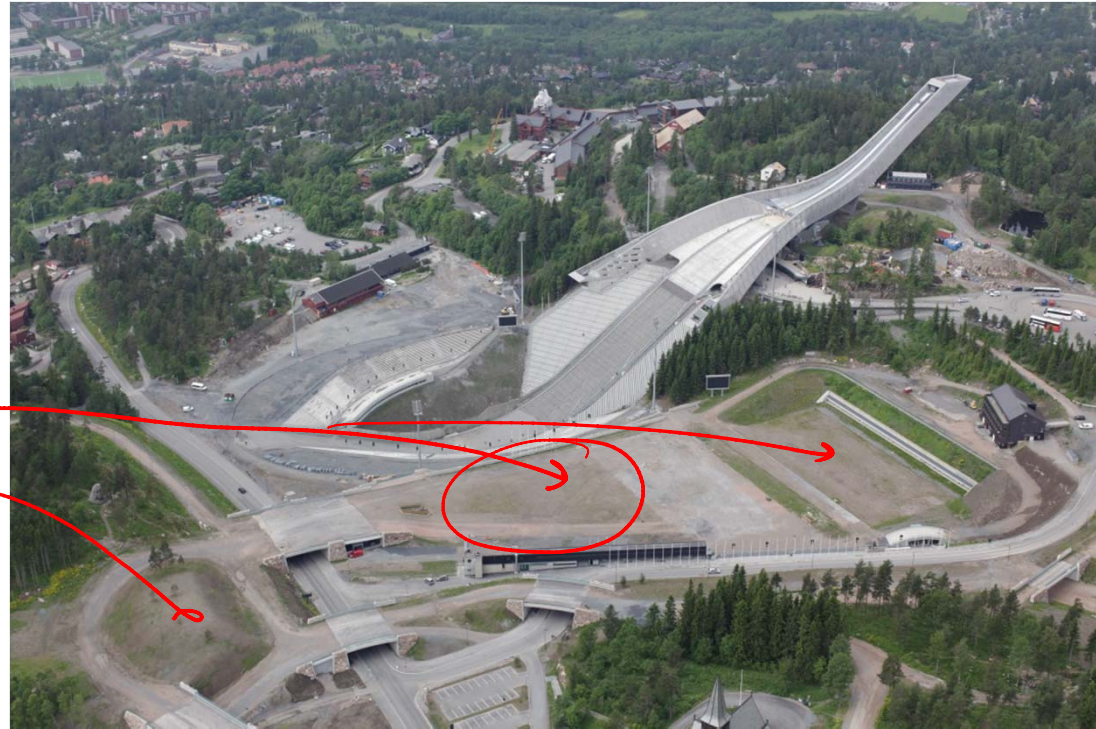
Step 3: Align images

Characteristics of good features



- Repeatability

Characteristics of good features



- Repeatability
- Distinctiveness

Characteristics of good features



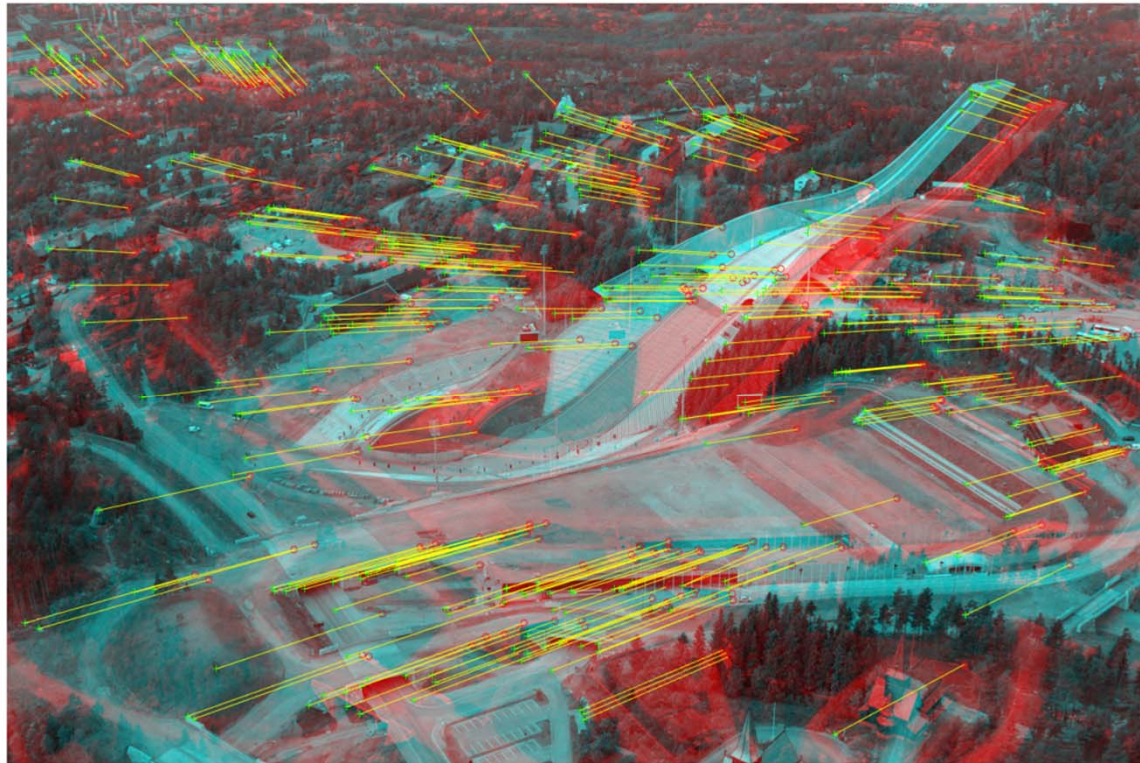
- Repeatability
- Distinctiveness
- Efficiency



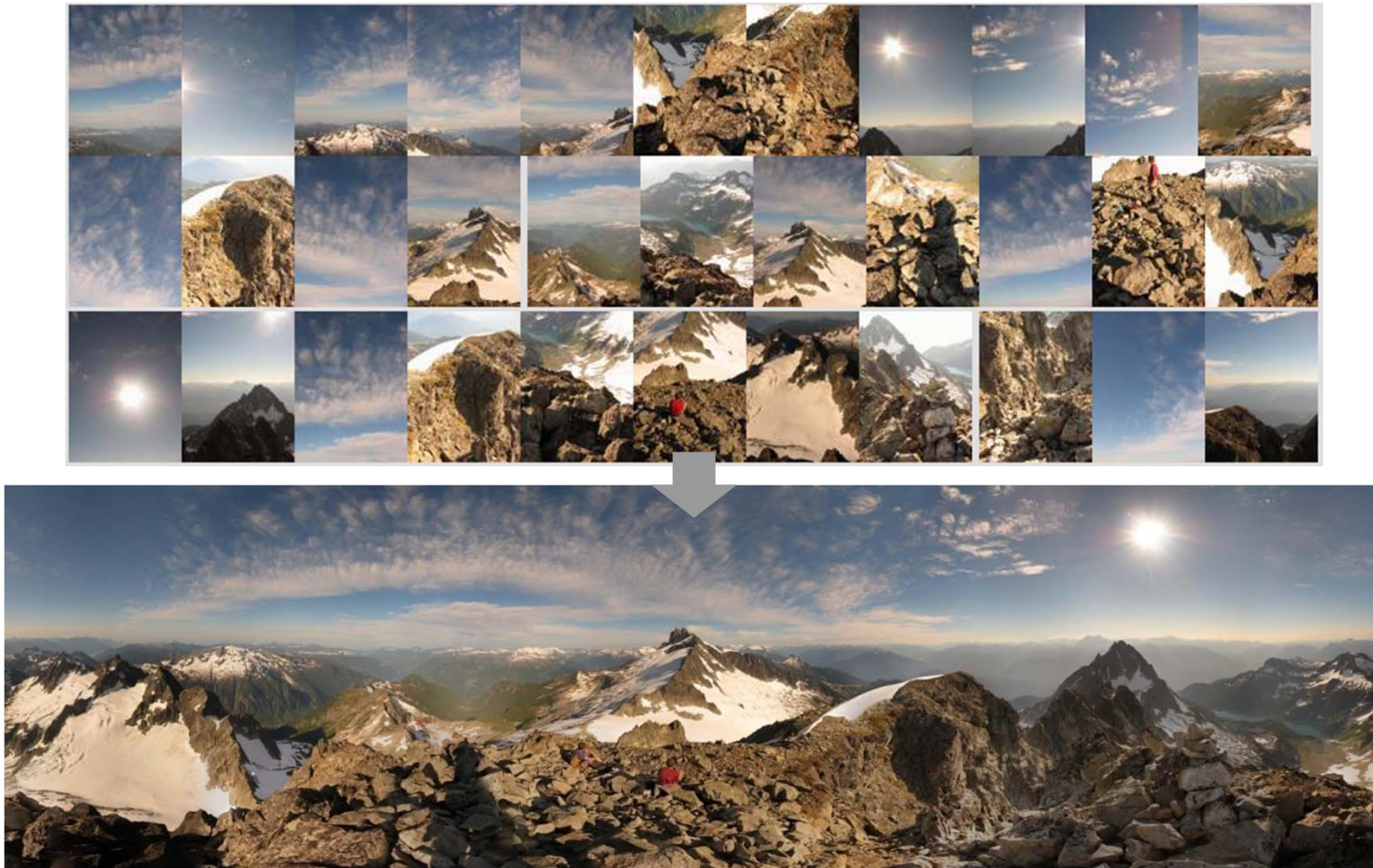
Correspondences across views



Correspondences across views

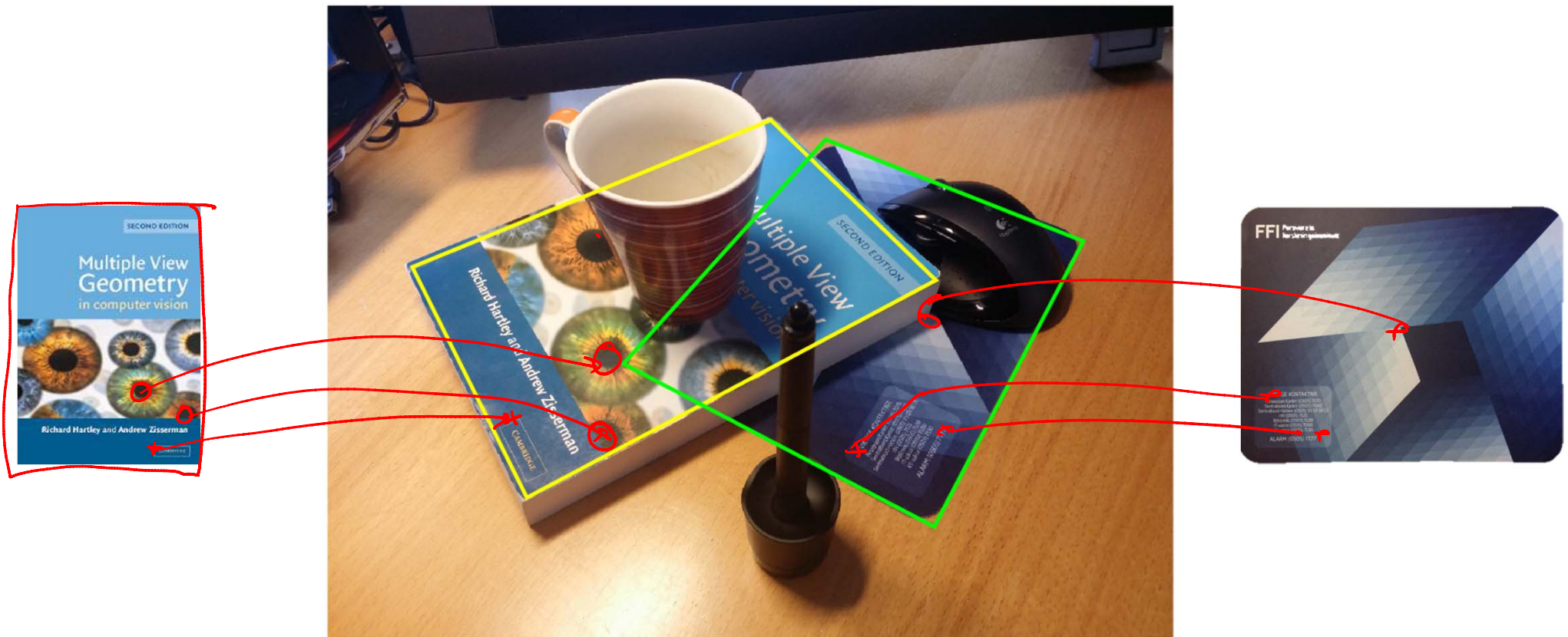


Applications: Automatic Panoramas

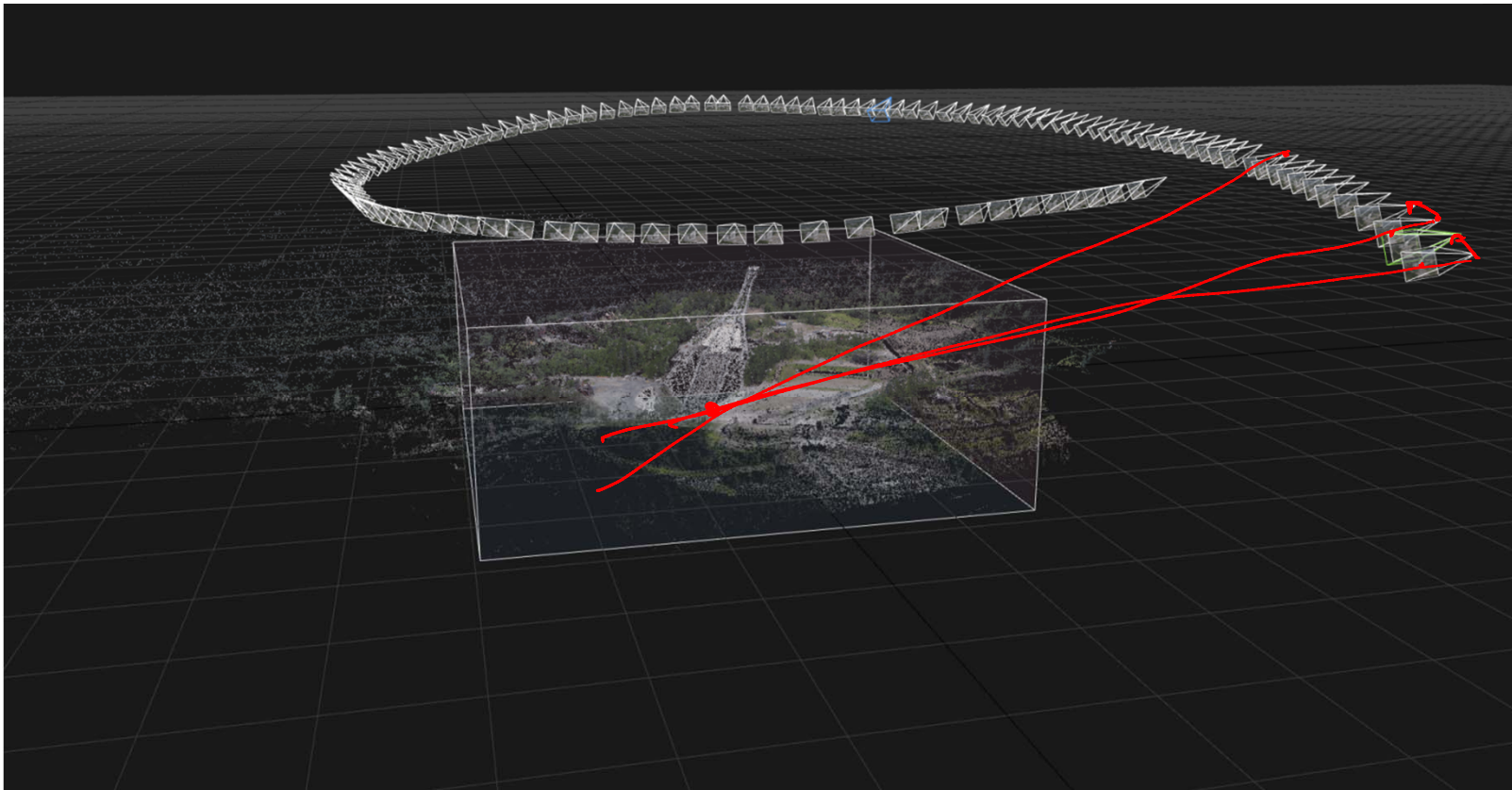


Credit: Matt Brown

Application: Object Recognition



Application: 3D reconstruction and navigation



Use of features in this course

- This week:
 - Lecture 3.2.1: Corner features
 - Lecture 3.2.2: Blob features
- Next week:
 - Feature descriptors
 - Feature matching
 - Homography estimation from correspondences
- Weeks after that:
 - Part II: World geometry and 3D
 - Part III: Scene analysis